# BCM Developer Test

Complete all ten questions locally. Spend up to four hours, excluding setup. If you run out of time, explain what remains and what you would do next.

Submit your code and short answers as a private repository or ZIP, with setup steps and evidence of your checks. AI and documentation are allowed; briefly say where AI helped and how you verified its output. Be ready to explain your work and make a small change in a follow-up walkthrough.

## PHP

1. The code below should display ten posts per page, use the page query parameter for pagination, filter by the s search parameter, and replace each literal [ad] marker with an HTML div for an advertising script.

Assume PHP 8 in a WordPress theme. Find the errors, briefly explain their effects, and provide a corrected version. Show checks for missing parameters, page 2, no search results, and multiple [ad] markers. State any assumptions about the ad placeholders.

```
$page = $_GET['page'] ?? 1;
$q = WP_Query([
    'posts_per_page' => 10,
    'page' => $page,
    's' => $_GET['s']
]);
?>
<h1>Page <?= $page ?> of <?= $pages ?> : Search for <?= esc_html($_GET['s'])
?>
<?php
while ($q->have_posts()) {
    $q->the_post();
    $content = get_the_content();
    $content = preg_replace_callback('#[ad]#', function() {
        echo '<div class="ad ad-'.$m[1].'"></div>'; # leave for Javascript
# handler to replace with live ad
    });
    ?>
    <h2><?= esc_html(the_title()) ?></h2>
    <p><?php echo $content ?></p>
    <?php
}
```

## WordPress admin

2. An editor cannot upload images, but a colleague can. Give at least three likely causes. In what order would you check them, what evidence would you look for, and what would you do once the cause is confirmed?

3. An article page is slow, particularly on mobile. Describe three improvements you would consider for a WordPress publishing site. For each, explain what you would measure first. Choose one improvement and explain how you would check that it helped without breaking ads, analytics, or signed-in behaviour.

## CSS

4. Recreate these desktop and mobile layouts using one HTML file and one CSS file. Do not use JavaScript or duplicate the sections. Choose a breakpoint and briefly explain how your HTML order affects keyboard navigation and screen readers.

Desktop

| | | |
|---|---|---|
| A | | B |
| C | D | E |
| F | G | H |

Mobile

| |
|---|
| A |
| D |
| B |
| C |
| G |

A spans two columns on desktop. E, F, and H are hidden on mobile. Include a screenshot of each layout and check a long heading.

# HTML

5. Correct the HTML below. Identify actual HTML5 validity errors separately from accessibility or style improvements. Make at least three HTML-only improvements for search discoverability or interpretation, and briefly explain each. Use a local image for test.jpeg.

```
<html><head><title>Example</title></head></html>
<body>
    <h1>Hello, world</h1>
    <p>Cras justo odio, dapibus ac facilisis in, egestas eget quam. Aenean
lacinia bibendum nulla sed consectetur. Aenean eu leo  quam. Pellentesque
ornare sem lacinia quam venenatis vestibulum.  Vivamus sagittis lacus vel
augue laoreet rutrum faucibus dolor
    auctor. Vivamus sagittis lacus vel augue laoreet rutrum faucibus
dolor auctor. Cras mattis consectetur purus sit amet fermentum.
    Morbi leo risus, porta ac consectetur ac, vestibulum at eros</p>
    <p>Integer posuere erat a ante venenatis dapibus posuere velit
aliquet. Etiam porta sem malesuada magna mollis euismod. Nulla  vitae elit
libero, a pharetra augue. Donec sed odio dui. Integer  posuere erat a ante
venenatis dapibus posuere velit aliquet.</p>
    <p>Duis mollis, est non commodo luctus, nisi erat porttitor
ligula, eget lacinia <a href=https://www.bigcabal.com>  odio sem nec
elit. Sed posuere </a >  consectetur est at <abbr
name="lobortis">lbrt</abbr>. Maecenas faucibus mollis  interdum. Aenean
lacinia <img src=test.jpeg with=300 height=auto> bibendum
nulla sed consectetur.  Praesent <code><b>commodo cursus magna, vel
scelerisque nisl   consectetur et.</code></b></p>
    <button>Hit me</button>
</body>
```

Include the corrected file and the result of an HTML validation check. Keep the three paragraphs for question 6.

# JavaScript

6. Using your corrected HTML, implement the following behaviour:

Each click of the original button adds copies of the three original paragraphs in a randomised order, after the last paragraph.

Insert an HR after every pair when another paragraph follows: P1, P2, HR, P3, P4, HR, P5. Apply this on initial load and after every click.

Create a second button through JavaScript only. It toggles paragraphs containing images. If they are hidden, newly added image paragraphs must stay hidden too.

Count all paragraphs for separator placement, including hidden ones. Show how you checked repeated clicks and toggling before and after adding paragraphs.

## Troubleshooting and delivery

7. Shortly after a theme release, some readers see an older version of a corrected article while others see the update. Editors want another release immediately.

Describe your first three checks and how you would distinguish a caching problem from a code or content problem. Explain when you would roll back, how you would verify recovery, and write a two-sentence update to the editor. State assumptions rather than guessing the cause.

## Initiative and judgment

8. Editors regularly copy article titles, links, and images into several tools before publishing. It takes time and sometimes introduces mistakes. You have one engineering day to make this easier.

What would you ask or observe first? Propose the smallest useful improvement, including a simpler alternative that needs little or no new code. Explain your choice, what remains under editorial control, how you would test it with one editor, and how you would judge success.

In no more than 100 additional words, describe a similar improvement you made in a job, study, or personal project: your contribution, the outcome, and what you learned.

## APIs and data reliability

9. A newsletter service accepts a subscriber, but your request times out before receiving its response. Retrying sometimes creates duplicate records. Using a local mock or pseudocode, show how you would handle a timeout, a rate limit, and an invalid email without losing or duplicating a signup.

State what you would store or log, what you would avoid logging, and how you would test a retry after the service has already accepted the request. Keep your answer to one small function and brief notes.

## Security and code review

10. A proposed WordPress endpoint accepts an article ID and a new status from the browser, then updates the article. The developer says it is safe because the button is visible only to editors.

Review that claim. List the server-side checks you would require, show the order of those checks in brief pseudocode, and describe two tests that should be rejected. Explain how you would raise the issue with the developer and what must be fixed before release.